

Exercise 23

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Introduction to Electrodynamics

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$$\rho = \frac{k}{x^2} \quad (a \leq x \leq b)$$

$$E(x) = \begin{cases} 0, & x < a \\ \frac{k(x-a)}{\epsilon_0 x^2}, & a < x < b \\ \frac{k(b-a)}{\epsilon_0 x^2}, & x > b \end{cases}$$

$$\begin{aligned} V_{center} &= - \int_{\infty}^b E(x) dx - \int_b^a E(x) dx - \int_a^0 E(x) dx \\ &= - \int_{\infty}^b \frac{k(b-a)}{\epsilon_0 x^2} dx - \int_b^a \frac{k(x-a)}{\epsilon_0 x^2} dx - \int_a^0 0 dx \\ &= \frac{k(b-a)}{\epsilon_0} \left. \frac{1}{x} \right|_{\infty}^b - \frac{k}{\epsilon_0} \left[\ln x \Big|_b^a + \frac{a}{x} \Big|_b^a \right] \end{aligned}$$

$$V_{center} = \frac{k}{\epsilon_0} \left[\left(1 - \frac{a}{b}\right) - \ln\left(\frac{a}{b}\right) - \left(1 - \frac{a}{b}\right) \right] = \frac{k}{\epsilon_0} \ln\left(\frac{b}{a}\right)$$

$$\frac{k}{\epsilon_0} \ln \frac{b}{a}$$

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